

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Reads graph and uses $10^{\log P}$ or $10^{\log V}$ to get either P or V	AO3.4	M1	$\log_{10} P = 2.18$ $P = 151$ $\log_{10} V = -0.15$ $V = 0.708$
	Correctly obtains P and V AWRT 150 and AWRT 0.71	AO1.1b	A1	
(b)	Calculates value of gradient to find d Condone use of $\log d =$ gradient Or uses a value of c plus a P/V pair to find d	AO3.4	M1	$\log_{10} P = \log_{10} c + d \log_{10} V$ Gradient = $d = -1.4$ Intercept = $\log_{10} c$ $c = 93.3$
	Obtains correct value for d AWRT -1.4 Not necessarily a decimal	AO1.1b	A1	
	Calculates value of intercept to find $\log_{10} c$ Or uses a value of d plus a P/V pair to find c	AO3.4	M1	
	Calculates correct value for c AWRT 93	AO1.1b	A1	
(c)	Uses their values of c and d in the formula $P = cV^d$	AO1.1a	M1	$P = 93.3 \times 2^{-1.4}$ $= 35.4$ kilopascals
	Obtains P value, including units AWFW 30 to 40	AO3.2a	A1	
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains (at least four) correct $\log_{10} y$ values, in	AO1.1a	M1	(1, 1.1) (2, 1.7) (3, 2.1) (4, 3.0)



	table or plotted			(5, 3.1) (6, 3.5)
	Plots all points correctly	AO1.1b	A1	(Points above plotted on grid)
(b)	Identifies $y = 1100$ and gives correct reason	AO2.2b	B1	(4, 1100), as it is not on the line that the other points are close to
(c)	Uses laws of logs. (May earn in part (a) if laws of logs were used there)	AO1.1a	M1	$\log_{10} y = \log_{10} k + x \log_{10} b$ Vertical intercept $c = 0.68$ ($= \log_{10} k$) Therefore from intercept: $k = 10^{0.68}$
	Draws straight line and calculates/measures the vertical intercept c and attempts 10^c or calculates/measures gradient m and attempts 10^m Alternatively uses regression line from calculator to get intercept and gradient	AO1.1a	M1	Gradient $m = 0.48 = \log_{10} b$ Therefore from gradient: $b = 10^{0.48}$
	Finds correct value of b from 'their' gradient, provided $0.45 < \text{'their' gradient} < 0.51$	AO1.1b	A1F	$k = 4.8$
	Finds correct value of k from 'their' intercept, provided $0.6 \leq \text{'their' intercept} \leq 0.8$	AO1.1b	A1F	$b = 3.0$
Total 7 marks				

Q3.

(a) $y = ax^n \Rightarrow \log_{10} y = \log_{10} ax^n$

$$\log_{10} y = \log_{10} a + \log_{10} x^n$$

Take logs and apply one log law in soln. correctly PI.

M1

$$\log_{10} y = \log_{10} a + n \log_{10} x$$

Apply a further log law correctly.

m1

$$Y = \log_{10} a + nX \text{ (which is a linear relationship between } Y \text{ and } X.)$$

Correct eqn. with base 10 (or lg or later evidence of use of base 10 if log without base here)

A1

3

(b) n = gradient of line

Stated or used.

Accept $n = \pm \frac{2}{3}$ OE as evidence

M1

$$n = -\frac{2}{3}$$

$$n = -\frac{2}{3} \text{ (OE 3sf)}$$

A1

$$\log_{10} a = 4$$

Equating c's constant term [must involve a log] in c's (a) eqn. to the Y-intercept value PI by correct value of a

M1

$$a = 10^4 (=10\,000)$$

A1

4

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Q4.

(a)

stretch	I	}	either order
SF 4	II		
in y-direction	III		
translate			
$\begin{pmatrix} e \\ 0 \end{pmatrix}$		}	

$$I + (II \text{ or } III)$$

M1A1

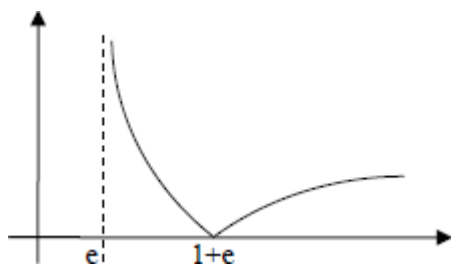
E1

accept 'e in positive x-direction'

B1

4

(b)



mod graph, in 2 connected sections, both in the first quadrant, touching x-axis

M1

curve touches x-axis at $1 + e$ (or 3.7 or better), and labelled (ignore scale)

A1

correct curvature, including at their $1 + e$, approx. asymptote at $x = e$

A1

3

(c) (i) $|4\ln(x - e)| = 4$

$4\ln(x - e) = 4$ $4\ln(x - e) = -4$ or better

must see 2 equations, condone omission of brackets

M1

(x =) $2e$ do not ISW
accept values of AWRT 5.42, 5.43, 5.44

A1

(x =) $e + e^{-1}$ or (x =) $e + \frac{1}{e}$ do not ISW
accept values of AWRT 3.08, 3.09
if M0 then $x = 2e$ with or without working scores SC1

A1

3

(ii) $x \geq 2e$

accept values of AWRT 5.42, 5.43, 5.44

B1

$$e < x \leq e + \frac{1}{e}$$

accept values of AWRT 2.72, 3.08, 3.09

if B2 not earned, then SC1 for any of

$$e \leq x \leq e + \frac{1}{e}, e < x < e + \frac{1}{e}, e \leq x < e + \frac{1}{e}$$

B2

3

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Q5.

(a) $\frac{1}{6}$ or $\left(\frac{1}{6}, 0\right)$

condone 0.167 AWRT

B1

1

(b) $\left(\frac{dy}{dx}\right) = \frac{1}{x}$

$$\frac{k}{x} \text{ where } k = 1, 6 \text{ or } \frac{1}{6}$$

$$k = 1$$

M1

A1

2

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Q6.

(a) Use of one law of logs or exponentials

M1

$$\lg a = c \text{ and } \lg b = m$$

OE; both needed

A1

$$\text{So } a = 10^c \text{ and } b = 10^m$$

A1

3

(b) Points (1, 1.08), (5, 1.43) plotted

M1 A0 if one point correct

M1A1

Straight line drawn through points
ft small inaccuracy

A1F

3

- (c) (i) Attempt at antilog of $Y(3)$
OE

M1

When $x = 3$, $Y \approx 1.25$ so $y \approx 18$
Allow AWRT 18

A1

2

- (ii) Attempt at a as antilog of Y -intercept
OE

M1

$a \approx 9.3$ to 10
AWRT

A1

2

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Q7.

- (a) $\lg(ab^x) = \lg a + \lg(b^x)$

Use of one log law

M1

$\dots = \lg a + x \lg b$

Use of another log law

M1

Correct relationship established

A1

3

[SC After M0M0, B2 for correct form]

- (b) (i) When $x = 2.3$, $Y \approx 1.1$, so $y \approx 12.6$
Allow 12.7; allow NMS

M1A1

(ii) When $y = 80$, $Y \approx 1.90$, so $x \approx 1.1$

M1 for $Y \approx 1.9$, allow NMS

M1A1

4

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Q8.

(a) $\lg y = \lg a + b \lg x$

M1 for use of one log law

M1A1

2

(b) Use of above result

M1

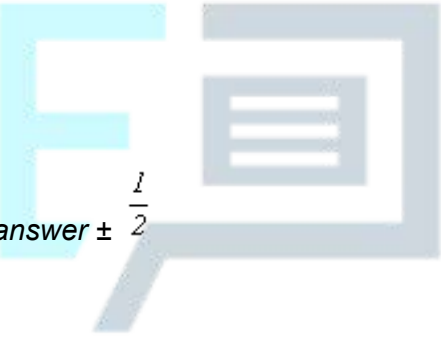
$a = 10$

A1

$b = \text{gradient}$

OE; PI by answer $\pm \frac{1}{2}$

m1


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A1

4

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Q9.

(a) Values 0.788, 0.992, 1.196 in table

B1 if one correct (or if wrong number of dp given)

B2,1

2

(b) $\lg ab^x = \lg a + \lg b^x$

M1

$\lg b^x = x \lg b$

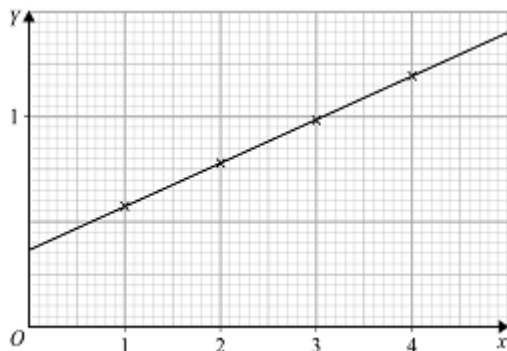
M1

So $Y = (\lg b)x + \lg a$
Allow NMS

A1

3

(c)



Four points plotted; ft wrong values in (a)

Good straight line drawn; ft incorrect points

B1F

B1F

2

(d) a = antilog of y -intercept
Accept 2.23 to 2.52

M1A1

b = antilog of gradient
Accept 1.58 to 1.62

M1A1

4

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Q10.

(a) X values 1.23, 2.18

Y values 0.70, 1.48

-1 for each error

B3,2,1

3

(b) $\lg y = \lg k + \lg x^n$

M1

$$\lg x^n = n \lg x$$

M1

$$\text{So } Y = nX + \lg k$$

A1

3

(c) Four points plotted

*B1 if one error here;
ft wrong values in (a)*

B2,1ft

Good straight line drawn

ft incorrect points (approx collinear)

B1ft

3

(d) Method for gradient

M1

Estimate for n

Allow AWRT 0.75 – 0.78; ft grad of candidate's graph

A1ft

2

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